

Differential Equations - Notes

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Part I

Basic Concepts

1 Definitions

1.1 What is a differential equation?

- A differential equation is an equation that contains derivatives, either ordinary or partial derivatives
- A popular example is newtons second law of motion, $F = ma$
- This is a differential equation since $a = \frac{dv}{dt}$
- Given this, we can actually write Newton's second law as a differential equation: $m \frac{dv}{dt} = F(t, v)$

1.2 What is Order?

- The order is the largest derivative present in the differential equation.
- For example, $\frac{\partial^3 u}{\partial x^2 \partial t} = 1 + \frac{\partial u}{\partial y}$ is a third order differential equation since it's greatest derivative is 3.

1.3 Ordinary vs. Partial Differential Equation

- An ordinary differential equation only has original differential equations, ode.
- A partial differential equation will have partial derivatives in it.

1.4 Linear Differential Equation

A linear differential equation is a differential equation following the form:

$$a_n(t)y^{(n)}(t) + a_{n-1}(t)y^{(n-1)} + \dots + a_1(t)y'(t) + a_0(t)y(t) = g(t)$$

1.5 Initial Conditions

An initial condition allows us to determine which solution we're after. For example, $y(t_0) = y_0$ or $y^{(k)}(t_0) = y_k$.

1.6 Implicit vs. Explicit Solutions

An implicit solution is a solution given in the form of $y = y(t)$

An explicit solution is a solution that isn't implicit.

Part II

First Order Differential Equations

2 Linear Differential Equations

2.1 Introduction

A linear first order differential equation is an equation in the form:

$$\frac{dy}{dt} + p(t)y = g(t)$$

- Here, we can assume $p(t)$ and $g(t)$ are continuous.

Now, let's assume the existence of a variable, $\mu(t)$, where we multiple all factors by it

$$\mu(t)\frac{dy}{dt} + \mu(t)p(t)y = \mu(t)g(t)$$

To solve problems using this:

- Put the differential equation in the correct form above.
- Find the integrating factor, $\mu(t)$, where $\mu(t) = e^{\int p(t)dt}$
- Multiply everything in the differential equation with $\mu(t)$, and verify the leftside becomes the product rule, $(\mu(t)y(t))'$.
- Integrate both sides.
- Solve for $y(t)$.

2.2 Example 1

Let's try to find the solution to the following differential equation:

$$\frac{dv}{dt} = 9.8 - 0.196v$$

We can see that we need to put the equation into linear form.

$$\frac{dv}{dt} + 0.196v = 9.8$$

Let's solve for $\mu(t)$

$$\mu(t) = e^{\int 0.196 dt} = e^{0.196t}$$

Next, we can plug in $\mu(t)$ in all variables

$$e^{0.196t} \frac{dv}{dt} + e^{0.196t} 0.196v = 9.8e^{0.196t}$$

We can rearrange some numbers:

$$e^{0.196t} \frac{dv}{dt} + 0.196e^{0.196t}v = 9.8e^{0.196t}$$

this resembles the product rule, hence we can say:

$$(e^{0.196t}v)' = 9.8e^{0.196t}$$

Now, we can integrate:

$$\int (e^{0.196t}v)' dt = \int 9.8e^{0.196t} dt$$

$$e^{0.196t}v = 9.8 \int e^{0.196t} dt$$

for the right equation, we can use u sub

$$e^u dt, \text{ where } u = 0.196t, du = 0.196dt, \text{ and } dt = \frac{du}{0.196}$$

We can plug this into our equation

$$\begin{aligned} e^{0.196t}v + k &= 9.8 \cdot \frac{1}{0.196} \int e^u du \\ e^{0.196t}v + k &= 50 \int e^u du \\ e^{0.196t}v + k &= 50e^{0.196t} + c \end{aligned}$$

Finally, we can solve for y, or v in this context:

$$v = \frac{50e^{0.196t} + c}{e^{0.196t}}$$

3 Separable Equations

3.1 Introduction

A separable differential equation is an equation where you can separate the variables:

$$N(y) \frac{dy}{dx} = M(x)$$

To put it simply, all the ys need to be on one side, xs on the other. To solve such a differential equation, you'd integrate both sides.

$$\int N(y) \frac{dy}{dx} dx = \int M(x) dx$$

We can use substitution to get this instead:

$$\int N(u) du = \int M(x) dx$$

3.2 Example

Solve the following differential equation and determine the interval of validity for the solution.

$$\frac{dy}{dx} = 6y^2x, \text{ and } y(1) = \frac{1}{25}$$

The first step is separating the Xs and Ys

$$\begin{aligned} dx \cdot \frac{dy}{dx} &= 6y^2x \cdot dx \\ \frac{dy}{y^2} &= \frac{6y^2x \cdot dx}{y^2} \\ \frac{1}{y^2} dy &= 6x dx \end{aligned}$$

From here, we can do integration:

$$\begin{aligned} \int \frac{1}{y^2} dy &= \int 6x dx \\ \int y^{-2} dy &= 3x^2 + C \\ -y^{-1} &= 3x^2 + C \\ -\frac{1}{y} &= 3x^2 + C \end{aligned}$$

now, we can solve for y:

$$y = \frac{1}{3x^2 + C}$$

Finally, we can solve for C to get our differential equation. This is trivial, as we have $y(1) = \frac{1}{25}$

$$\frac{1}{25} = \frac{1}{3+C}$$

We can easily figure out that $C = 22$.

Hence our final differential equation is:

$$y = \frac{1}{3x^2 + 22}$$

4 Exact Equations

4.1 Introduction

An exact equation follows the form: $M(x, y) + N(x, y) \frac{dy}{dx} = 0$

Let's create two variables:

- $\psi_x = M(x, y)$
- $\psi_y = N(x, y)$

Plugging it into our equation, we get:

$$\psi_x + \phi_y \frac{dy}{dx} = 0$$

We can then use the chain rule to find the derivative of the differential equation:

$$\frac{d}{dx}(\psi(x, y(x))) = 0$$

this gives us an implicit solution: $\psi(x, y) = c$.

4.2 What is a exact equation

An exact equation is an equation, given M_y and N_x , if their partial derivatives are equal to eachother.

4.3 Example

Solve the following IVP and find the interval of validity for the solution

$$2xy - 9x^2 + (2y + x^2 + 1)\frac{dy}{dx} = 0, \text{ where } y(0) = -3.$$

- We can set two variables.
- $M(x, y) = 2xy - 9x^2$
- $N(x, y) = 2y + x^2 + 1$

The first thing we can do is check if this is an exact equation:

$$\begin{aligned}\frac{\partial M}{\partial y} &= 2x \\ \frac{\partial N}{\partial x} &= 2x\end{aligned}$$

We can see the partial derivatives are identical, hence it's an exact equation.
Next, we need to integrate one of the equations:

$$\int M(x, y) = \int 2xy - 9x^2 dx = x^2y - 3x^3 + h(y)$$

Notice how, instead of a constant, we use $h(y)$.

We actually have $h'(y)$, it's $N = 2y + x^2 + 1$.

Now, we find $\frac{\partial M}{\partial y}$ the integral of M, giving us $x^2 + h'(y)$

We can set $\frac{\partial M}{\partial y} = M$, or $x^2 + h'(y) = 2y + x^2 + 1$.

$$\begin{aligned}x^2 + h'(y) - x^2 &= 2y + x^2 + 1 - x^2 \\ h'(y) &= 2y + 1\end{aligned}$$

We can integrate $h'(y)$ to get $h(y)$.

$$\int 2y + 1 dy = y^2 + y + k$$

We can plug this into our equation:

$$\psi(x, y) = x^2y - 3x^3 + y^2 + y + k$$

Now, we need to solve for k, or c

$$\begin{aligned}\psi(x, y) &= x^2y - 3x^3 + y^2 + y + k = c \\ \psi(x, y) &= x^2y - 3x^3 + y^2 + y + k - k = c - k \\ \psi(x, y) &= x^2y - 3x^3 + y^2 + y = c \\ \psi(x, y) &= x^2y - 3x^3 + y^2 + y = c \text{ Here, we use } y(0) = -3 \\ \psi(x, y) &= (0)^2(-3) - 3(0)^3 + (-3)^2 + (-3) = c \text{ Here, we use } y(0) = -3 \\ \psi(x, y) &= 9 - 3 = c, \text{ where } c = 6 \\ \text{hence our solution is } \psi(x, y) &= x^2y - 3x^3 + y^2 + y - 6 = 0\end{aligned}$$

5 Bernoulli Differential Equation

5.1 Introduction

Bernoulli Equations are equations that follow the form:

$$y' + p(x)y = q(x)y^n$$

Here, we assume $p(x)$ and $q(x)$ are continuous functions, and n is a number. if $n = 0$ or 1 , we'll have a linear equation, which is easy to solve.

if n isn't 0 or 1 , we need to divide both sides by y^n , substitute $v = y^{1-n}$. Once we do this, we'll get a linear equation we can solve.

5.2 Example

Solve the following IVP and find the interval of validity for the solution:

$$y' + \frac{4}{x}y = x^3y^2, \text{ where } y(2) = -1 \text{ and } x > 0$$

We can see our differential equation resembles the bernoulli equation, where:

- $p(x) = \frac{4}{x}$
- $q(x) = x^3$
- $n = 2$

now, we need to divide both sides by y^n

$$\begin{aligned}\frac{y' + \frac{4}{x}y}{y^2} &= \frac{x^3y^2}{y^2} \\ \frac{y' + \frac{4}{x}y}{y^2} &= \frac{x^3y^2}{y^2} \\ y^{-2}y' + \frac{4}{x}y^{-1} &= x^3\end{aligned}$$

From here, we can substitute, using $v = y^{1-n} = y^{1-2} = y^{-1}$ and $v' = -y^{-2}y'$, giving us:

$$-v' + \frac{4}{x}v = x^3$$

Because $\frac{dv}{dt}$ needs to be positive, we need to multiply both sides by -1 to get:

$$v' - \frac{4}{x}v = -x^3$$

Next, we need to multiply all variables by $\mu(x)$

$$\mu(x)v' - \mu(x)\frac{4}{x}v = -\mu(x)x^3$$

next, we need to find $\mu(x)$

$$\mu(x) = e^{\int -\frac{4}{x} = -4 \int \frac{1}{x} = -4 \ln|x| \rightarrow \ln|x|^{-4}} = x^{-4}$$

We can plug this in to get

$$x^{-4}v' - x^{-4}\frac{4}{x}v = -x^{-4}x^3 = -x^4v' - 4x^{-5}v = -x^{-1}$$

You can see that this resembles the product rule, hence we can simplify this equation to:

$$-x^{-4}x^3 = -x^4v' - 4x^{-5}v = (x^{-4}v)'$$

We can then integrate:

$$\begin{aligned} \int (x^{-4}v)' &= \int -x^{-1}dx \\ x^{-4}v &= -\ln|x| + c \\ v(x) &= cx^4 - x^4\ln x, \text{ or } v = cx^4 - x^4\ln x \end{aligned}$$

We can then solve for c by using the condition $y(2) = -1$. We can also plug y^{-1} back in

$$\begin{aligned} y^{-1} &= cx^4 - x^4\ln x \\ -1^{-1} &= c2^4 - 2^4\ln 2 \\ -1 + 16\ln 2 &= 16c - 16\ln 2 + 16\ln 2 \\ \frac{-1+16\ln 2}{16} &= \frac{16c}{16} \quad c = \frac{-1+16\ln 2}{16} \end{aligned}$$

We can solve for y, bla bla bla.

6 Substitutions

6.1 Introduction

In addition to Bernoulli's Differential equation, there's also another one known as homogenous differential equations.

6.1.1 Homogenous equations

Homogenous equations are equations that take the form:

$$y' = F\left(\frac{y}{x}\right)$$

If this is the case, we can use the following substitutions:

$$\begin{aligned} v &= \frac{y}{x}, \text{ which can be written as } y = xv \\ y' &= v + xv' \end{aligned}$$

using this will give us a separable differential equation, which we can easily solve.

Example Solve the following IVP and find the interval of validity for the solution:

$$xyy' + 4x^2 + y^2 = 0, \text{ where } y(2) = -7 \text{ and } x > 0$$

We can move $4x^2 + y^2$ to the other side of the equation, giving us:

$$xyy' = -4x^2 - y^2$$

we can then divide both sides by two:

$$\frac{y}{x}y' = -4 - \frac{y^2}{x^2} = -4 - \left(\frac{y}{x}\right)^2$$

Hence, we proved our equation here is homogenous. Next, we can apply the substitution here.

$$v(v + xv') = -4 - v^2$$

We can then separate variables, then use integration to solve our differential equation.

$$\int \frac{v}{2v^2} dv = \int -\frac{1}{x} dx$$

6.1.2 Other equation

If you have a differential equation with the form: $y' = G(ax + by)$, then you can apply the following substitution:

$$\begin{aligned} v &= ax + by \\ v' &= a + by' \end{aligned}$$

7 Intervals of Validity

7.1 Theorem one

When it comes to intervals of validity, there's an important idea:

$$y' + p(t)y = g(t), \text{ where } y(t_0) = y_0$$

If $p(t)$ and $g(t)$ are continuous functions on an open interval $\alpha < t < \beta$, containing an interval t_0 , then there's a unique solution to the IVP on the interval.

7.1.1 Example

Let's try to determine the intervals of validity for the following initial value problem.

$$(t^2 - 9)y' + 2y = \ln|20 - 4t|, \text{ where } y(4) = -3$$

Let's start off by dividing both sides by $(t^2 - 9)$, giving us:

$$y' + \frac{2}{t^2-9}y = \frac{\ln|20-4t|}{t^2-9}$$

In order to find non continuous intervals, we need to find points that aren't defined. In this example, dividing by zero, or if you get $\ln|0|$.

$$20 - 4t = 0 \rightarrow t = 5$$

$$t^2 - 9 = 0 \rightarrow (t - 3)(t + 3) = 0 \rightarrow t = \pm 3$$

hence, our Intervals of Validity include:

$$-\infty < t < -3, -3 < t < 3, 3 < t < 5, 5 < t < \infty$$

7.2 Theorem 2

Let's consider the following IVP:

$$y' = f(t, y), \text{ where } y(t_0) = y_0$$

if $f(t, y)$ and $\frac{\partial f}{\partial y}$ are continuous functions in some rectangle $\alpha < t < \beta$, and $\gamma < y < \delta$, containing the point (t_0, y_0) , then there's a unique solution to the IVP in some interval $t_0 - h < t < t_0 + h$

7.2.1 Example

Determine the interval of validity for the initial value problem below and give its dependence on the value of y_0

$$y' = y^2, \text{ where } y(0) = y_0$$

Here, we can define the variables:

- $f(t, y) = y^2$
- $t_0 = 0$
- y_0 is the initial value.

We first need to check for continuity for $f(t, y)$ and $\frac{\partial f}{\partial y}$

- $f(t, y) = y^2$ is continuous
- $\frac{\partial f}{\partial y} = 2y$, which is also continuous.

Our equation also contains the point (t_0, y_0) , so we're good. we can now now, we can work on this:

$$\begin{aligned} \frac{dy}{dt} &= y^2 \\ dy &= y^2 dt \\ \frac{1}{y^2} dy &= dt \\ \int \frac{1}{y^2} dy &= \int dt \\ -\frac{1}{y} &= t + c \end{aligned}$$

We're given $y(0) = y_0$, so we can figure out c with it.

$$-\frac{1}{y_0} = c$$

therefore, our solution is $-\frac{1}{y} = t - \frac{1}{y_0}$.

We can rearrange the variables to get $y = \frac{y_0}{1 - y_0 t}$

We can then find out that y is undefined when $t = \frac{1}{y_0}$.